

Word Problems & Reasoning



Number and digit reasoning

- 1 A number decreased by 9 equals 34. Find the number.
 - 2 Three times a number, minus 7, equals 32. Find the number.
 - 3 The sum of two numbers is 58. One number is 12 more than the other. Find both numbers.
 - 4 A two-digit number has digit sum 11. Reversing its digits gives a number 45 more than the original. Find the original number.
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Age problems

- 5 A father is 4 times as old as his daughter. In 5 years, he will be 3 times as old as she will be. Find their current ages.
 - 6 Sara is twice as old as her brother. In 6 years, the sum of their ages will be 42. Find their current ages.
 - 7 Ahmed is 5 years older than Salem. Seven years ago, Ahmed was twice as old as Salem. Find their current ages.
 - 8 The sum of a mother's and daughter's ages is 50. The mother is 20 years older than the daughter. Find both ages.
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Distance, rate and time

- 9 A train travels 180 km in 2 hours. At the same speed, how far does it travel in 5 hours?
 - 10 Two cities are 350 km apart. A car leaves one city for the other at 70 km/h. How long does the trip take?
 - 11 Two cyclists start at the same point and ride in opposite directions, at 15 km/h and 20 km/h. After how many hours will they be 105 km apart?
 - 12 A plane flies against a headwind at 480 km/h and covers 1,440 km. How many hours does the flight take?
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Work-rate problems

- 13 Worker A can paint a room in 6 hours; Worker B can paint the same room in 3 hours. Working together, how long will it take them?
- 14 A pipe fills a tank in 10 hours; a second pipe fills it in 15 hours. Working together, how long to fill the tank?
- 15 6 workers can complete a job in 8 days. How many days would 4 workers take, at the same rate?
- 16 A pipe fills a tank in 12 hours, but a leak can empty a full tank in 24 hours. If both act at once, how long to fill the tank?
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Mixture and money problems

- 17 A jeweler mixes gold worth 200 riyals/gram with silver worth 50 riyals/gram to make 10 grams of a mixture worth 110 riyals/gram. How many grams of gold are used?
- 18 A shopkeeper has 15 riyals in 1-riyal and 2-riyal coins, 10 coins in total. How many of each coin does she have?
- 19 A solution is 40% acid. How many liters of pure water must be added to 20 liters of this solution to dilute it to 25% acid?
- 20 Event tickets cost 30 riyals for adults and 18 riyals for children. 150 tickets were sold for 3,780 riyals in total. How many adult tickets were sold?
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Logical deduction

- 21 Ali, Basim and Khalid have 36 books together. Ali has twice as many as Basim, and Khalid has 6 more than Ali. Find how many books each has.
- 22 A number between 50 and 70 is divisible by both 3 and 4, but not by 8. Find the number.
- 23 In a class, every student plays football or basketball (or both). 25 play football, 18 play basketball, and 10 play both. How many students are in the class?
- 24 Four consecutive even integers sum to 92. Find the largest one.
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Working backward

- 25 After spending half his money on a book, then 20 riyals on lunch, Yousef has 30 riyals left. How much did he start with?

- 26 A number is tripled, then 8 is subtracted, giving 25. Find the original number.
- 27 After a 15% pay cut, an employee's salary is 3,400 riyals. Find the salary before the cut.
- 28 A water tank is $\frac{1}{4}$ full. After adding 60 liters, it becomes $\frac{5}{8}$ full. Find the tank's total capacity.
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Mixed word problems

- 29 A rectangular field is twice as long as it is wide, and its perimeter is 90 m. Find its length and width.
- 30 Layla buys 3 notebooks and 2 pens for 29 riyals. Fahad buys 1 notebook and 4 pens for 23 riyals, at the same prices. Find the price of one notebook and one pen.
- 31 A car rental costs 80 riyals per day plus 0.50 riyals per km driven. A 3-day rental cost 435 riyals in total. How many km were driven?
- 32 The sum of three consecutive odd integers is 63. Find the integers.
- 33 A shop sells notebooks for 6 riyals each, with a "buy 3, get 1 free" offer. Layla needs 20 notebooks. How much will she pay?
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More age and number problems

- 34 The sum of a number and its square is 42. Find the positive value of the number.
- 35 A number's triple exceeds its double by 15. Find the number.
- 36 In 10 years, Noura will be 3 times as old as she was 2 years ago. Find her current age.
- 37 The sum of two consecutive multiples of 5 is 75. Find the larger one.
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More rate and proportion problems

- 38 A photocopier prints 45 pages per minute. How many minutes does it take to print 810 pages?
- 39 A recipe for 6 people uses 450 g of rice. How much rice is needed for 15 people, at the same rate per person?

- 40** A factory produces 1,200 items in 8 hours. At the same rate, how many items would it produce in a 10-hour shift?
- 41** Three taps together fill a pool in 4 hours. Two of the taps alone would take 6 hours. How long would the third tap alone take to fill the pool?